Midterm 1

Graded

Student

Amina Uzma

Total Points

62 / 100 pts

Question 1

Question 1 — **9** / 30 pts

1.1 **a** — **9** / 15 pts

- **– 0 pts** Correct
- **– 3 pts** first one is wrong
- ✔ **– 3 pts** second is wrong
- **– 3 pts** third is wrong
- **– 3 pts** fourth is wrong
- ✔ **– 3 pts** fifth is wrong

1.2 **b** — **0** / 7 pts

- **– 0 pts** Correct
- **– 1 pt** unit is wrong (should be 6000 kJ)
- **– 1 pt** minutes to seconds coversion
- **– 1 pt** watts= joules/seconds
- ✔ **– 7 pts** not done or wrong

1.3 **c** — **0** / 8 pts

- **– 0 pts** Correct
- **– 0 pts** Pressure = rho*g*h
- ✔ **– 8 pts** not done or wrong
- **– 1 pt** unit is wrong
- **– 1 pt** rho is wrong
- **– 1 pt** needs to determine the difference

Question 2

Question 2 **35** / 35 pts

2.1 a **2** / 2 pts

✔ **− 0 pts** Correct

2.2 b **4** / 4 pts

✔ **− 0 pts** Correct

− 1 pt 3-1 is wrong

− 4 pts wrong or not done

− 1 pt not water, this is air. Not dome drawing

− 1 pt x axis should be specific volume, not volume.

2.3 c,d **5** / 5 pts

✔ **− 0 pts** Correct

− 1 pt closed system

− 1 pt deltaU = Q-W

2.4 e **15** / 15 pts

✔ **− 0 pts** Correct

− 5 pts W1-2 is wrong

− 4 pts W2-3 is wrong

− 6 pts W3-1 is wrong

− 1 pt W1-2 should be negative

− 2 pts n value for polytropic process is wrong

− 2 pts unit conversion

− 1 pt W3-1 should be positive

− 1 pt W3-1 unit conversion

− 3 pts W3-1 should be calculated by natural log

− 1 pt W 3-1 calculation error

− 1 pt calculation error W1-2

2.5 f **9** / 9 pts

✔ **− 0 pts** Correct

− 1 pt calculation error

− 9 pts not done or wrong

Question 3

Question 3 **18** / 35 pts

3.1 a **2** / 2 pts

- ✔ **− 0 pts** Correct

3.2 b **2** / 6 pts

- **− 0 pts** Correct
- **− 3 pts** domes are missing on the graphs
- **− 5 pts** states are not labeled on the graph
- **− 1 pt** state 1 is wrong on Pv graph
- ✔ **− 1 pt** state 1 is wrong on Tv graph
- **− 1 pt** state 2 is wrong on Pv graph
- ✔ **− 1 pt** state 2 is wrong on Tv graph
- ✔ **− 1 pt** state 3 is wrong on Pv graph
- ✔ **− 1 pt** state 3 is wrong on Tv graph

3.3 c,d **5** / 5 pts

- ✔ **− 0 pts** Correct
- **− 5 pts** not done or wrong
- **− 1 pt** closed system

3.4 e **9** / 10 pts

- **− 0 pts** Correct
- **− 10 pts** not done or wrong
- **− 6 pts** W2-3 is not done or wrong (should be constant P)
- ✔ **− 1 pt** calculation error
- **− 1 pt** W2-3 should be negative
- **− 1 pt** P is wrong
- **− 2 pts** v3 is wrong
- **− 2 pts** v2 is wrong

3.5 **f** **0** / 10 pts

- **− 0 pts** Correct
- **− 1 pt** calc errro
- **− 3 pts** x not calculated or wrong
- **− 3 pts** u1 is wrong
- ✔ **− 10 pts** not done
- **− 1 pt** u3 is wrong
- **− 2 pts** deltaU calculations are wrong

3.6 **g** **0** / 2 pts

- **− 0 pts** Correct
- **− 1 pt** state 1 is wrong
- **− 1 pt** state 2 is wrong
- ✔ **− 2 pts** not done or wrong
- **− 0.5 pts** state 3 is wrong

CHE 201 MidTerm Exam 1 September 29th, 2025

NAME: Amina Ikram

Please fill in answers in the space supplied and show all your work to receive full credit.

1. **a) (15 pts)** State whether the following statements are **True of False**

F An ideal gas at a given state expanded to a fixed final volume at constant pressure or at constant temperature. Value of work is smaller in constant pressure case.

F Seven people stand on a scale and have a mass of 1280 lb. A specific property called the pep is defined as the number of people divided by mass. Pep is an intensive property.

F Isochoric process means no heat transfer.

F A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality more than 0.5.

mass vapor / mass $\frac{1}{6}$

T Energy transferred by heat to a closed system can be calculated by the area under the curve in P-V diagram.

b) (7pts) (FE Exam)
A 2-kW electric resistance heater in a room is turned on and kept on for 50 min. Determine the amount of energy transferred.

50 min

$KE = \frac{1}{2}mv^2$

$W = PV$

c) (8pts) (FE Exam)
Consider a fish swimming 5 m below the free surface of water. Determine the increase in the pressure exerted on the fish when it dives to a depth of 25 m below the free surface.

5 ↓ 25

mgh $m(9.8)(-20)$

$W = PV$

2. (35 points)

A gas in a piston-cylinder assembly (7kg) at 5 bar and 15 m^3 undergoes a cycle with three processes in series.

Process 1-2: Isobaric compression to 10 m^3.

Process 2-3: Isochoric increase of pressure to 7.5 bar.

Process 3-1: Polytropic expansion to initial state.

Kinetic and potential effects can be ignored.

a) (2 pts) Draw a schematic of the system and label the system boundary. ✓

b) (4 pts) Carefully sketch the process on a P-v diagram (Pressure-specific volume) ✓

c) (3 pts) State the engineering model ✓

d) (2 pts) Write the simplified energy balance ✓

e) (15 pts) Calculate the work transferred in kJ for each step. Is work done on the system or by the system? ✓

f) (9 pts) Calculate the overall heat transferred in kJ. Is heat transferred to the system or from the system? ✓

2) a)

Process 1-2

constant Pressure

m=7kg m=7kg

P=5bar P=5bar

V=15m3 10m3

Process 2-3

constant volume

m=7kg m=7kg

P=5bar P=7.5bar

V=10m3 V=10m3

Process 3-1

PV^n=const

m=7kg m=7kg

P=7.5bar P=5bar

V=10m3 V=15m3

c) $\Delta KE = \Delta PE = 0$

Closed system

d) $\Delta E = Q - W$

$\Delta E = \Delta KE + \Delta PE + \Delta U$

$\Delta U = Q - W$

e) $W = \int_{V_1}^{V_2} P\,dv$

Process 1-2

$W = P(V_2 - V_1)$

$W = 5(10-15)$

$W = -2500$ kJ work done on system (−)

Process 2-3

$W = \int_{V_1}^{V_2} P\,dv$

$W = \int_{10}^{10} P\,dv$

$\Delta V = 0$ so Work = 0

& constant volume

Process 3-1

$PV^n = PV^n$

$(7.5)(10)^n = (5)(15)^n$

$\frac{10^n}{15^n} = \frac{5}{7.5}$

$\left(\frac{2}{3}\right)^n = \frac{2}{3}$

$n=1$

$P = \frac{75}{V1} = V^{-1}$

$W = \int_{10}^{15} V^{-1}\,dv$

$W = 75\ \ln(V)\Big|_{10}^{15}$

$W = 75(\ln(15) - \ln(10))$

$W = 3041$ kJ

work done by the system (+)

f) $\Delta E = Q - W$

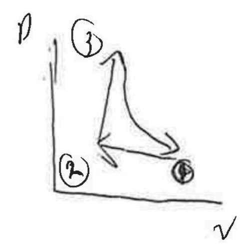
$\Delta E = 0$ in thermodynamic cycle

$Q = W$

$Q = W = 3041 - 2500$

$= 541$ kJ

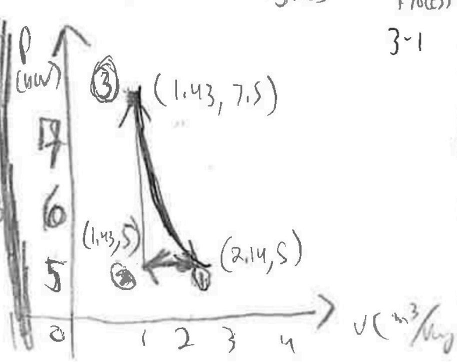
heat transferred to the system



b)

$v_1 = \frac{15m^3}{7kg} = 2.14$

$v_2 = v_3 = \frac{10m^3}{7kg} = 1.43$



3. (35 points)

5kg of liquid-vapor mixture of water contained in a piston–cylinder assembly undergoes a process from an initial state where the temperature is 90°C.

- **Process 1–2** System is heated at constant volume to a saturated vapor state at 170°C.
- **Process 2-3** System is cooled at constant pressure to 120 °C.

Kinetic and potential effects can be ignored.

a) (2 pts) Draw a schematic of the system and label the system boundary.
b) (6 pts) Carefully sketch the process on a:
(1) P-v diagram (Pressure-specific volume)
(2) T-v diagram (Temperature-specific volume)
c) (3 pts) State the engineering model
d) (2 pts) Write the simplified energy balance
e) (10 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
f) (10 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
g) (2 pts) Calculate the mass of vapor in each state in kg.

3 (a) Process 1-2

constant volume

m = 5kg → m = 5kg

T_1 = 90°C → T_2 = 170°C

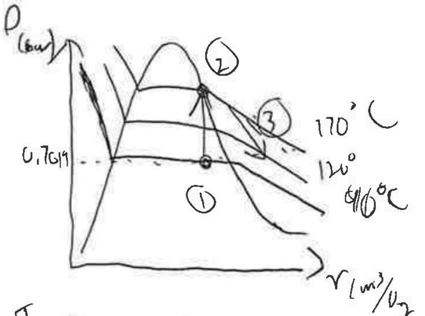
Process 2-3

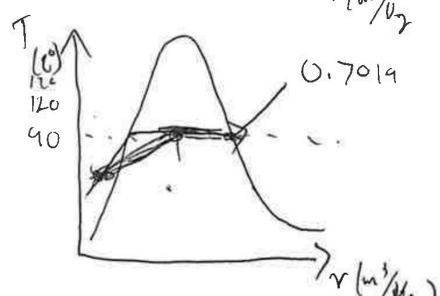
constant pressure

m = 5kg → m = 5kg

T_2 = 170°C → T_3 = 120°C

(b)





(c) closed system

$\Delta KE = \Delta PE = 0$

(d) $\Delta E = \cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U$

$\Delta E = Q - W$

$\Delta U = Q - W$

	1	2	3
Phase	liquid vapor mix	Saturated vapor	
T (°C)	90°	170°	120°
P (bar)	0.7019	7.917	
u (kJ/kg)		2576.5	
v (m³/kg)		0.2428	

(e) $W = \int_{v_1}^{v_2} P\,dv$

Process 1-2: $W = 0$ → constant volume

Process 2-3: $W = \int_{v_1}^{v_2} P\,dv$

$V_2 = 0.2428 \times 5 = 1.214\ m^3$

$V_1 = 1.0360 \times 10^{-3} \times 5 = 0.00518\ m^3$

$W = 7.917(1.214 - 0.00518)$

$W = 121$ kJ work done by system

(f) $\Delta U = Q - W$

$Q = W + \Delta U$

$Q = 121 + 5($

(g) $X = \frac{m_{vapor}}{mass\ total}$

State 2 = 0 kg (vapor)

State 3 =